Reporting Data Science Results

Why reporting matters

Reporting is turning analysis into a message that the intended reader can understand, trust and act on.

1 Results must be understood

A good model with a poor report rarely changes a decision.

2 Trust is earned in the details

Readers look for evidence and limits before they believe a number.

3 Work must be checkable

Others should be able to rerun it and get the same result.

Principle: a report is a set of claims with evidence. For every claim, show the evidence and how strong it is.

Today's focus: data storytelling for non-technical readers, and honest, ethical charts.

Know your audience

An audience analysis identifies who will read the results, what decision they face, and how much technical detail they need.

Reader	Wants to know	Format
Technical peers	Methods, evaluation design, how to reproduce	Full report, appendix, code
Domain experts (clinicians, managers)	Does it work for my cases? What are the risks?	Short report, plain language, key figures
Decision makers	What should we do, and how sure are we?	One-page summary with a recommendation
Future maintainers	Why was it done this way? How do I rerun it?	README, data dictionary, comments

Rule: lead with what the reader cares about; put technical detail where only those who want it will look.

What is data storytelling?

Data storytelling is explaining what data means, and what to do about it, through a clear narrative supported by visuals.

1 Data

The evidence.

2 Narrative

What it means and what to do.

3 Visuals

A chart that makes the point obvious.

Not spin: the story must be the one the data supports, including what it does not show.

The story arc

A story arc is the order in which a message unfolds: context, problem, insight, action. Each stage answers one question the audience has.

1 Context	2 Problem	3 Insight	4 Action
Why should I listen?	**What is at stake?**	**What did you find?**	**What should we do?**
1 in 5 heart-failure patients is readmitted within 30 days.	Each readmission costs about $15,000.	The 20% flagged by the model are half of all readmissions.	Call flagged patients within 48 hours; pilot on two wards.
Start from what they know.	*Cost, risk or opportunity.*	*One finding, in their units.*	*A step, an owner, a date.*

One insight per story: if there are three findings, choose the one that drives the action.

Big idea: “Calling the 20% highest-risk patients within 48 hours could prevent many readmissions.”

The story arc in action

Ordering by the arc means presenting results in the order the audience needs them, not in the order the analysis was done.

Analyst's order (chronological)

- We collected 3 years of hospital records.
- We cleaned the data and engineered 40 features.
- We compared 5 models; gradient boosting was best.
- Recall was 0.85 at a threshold of 0.4.
- *Maybe the wards could use it?*

Audience's order (story arc)

- **Context:** 1 in 5 heart-failure patients returns within 30 days.
- **Problem:** each return costs about $15,000.
- **Insight:** the patients we flag are half of all returns.
- **Action:** call them within 48 hours; pilot on two wards.
- *Methods: in the appendix.*

Order for the audience: lead with why it matters and what to do; move methods to the end or an appendix.

Translating model outputs into plain language

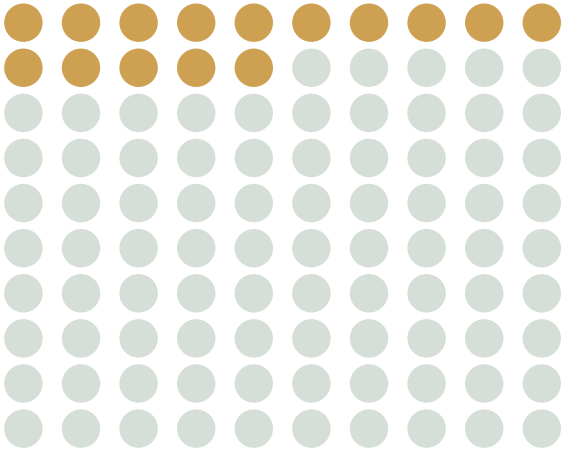
Translation means restating a technical result in the reader's terms: what it means for their cases, their time or their money.

Model output	Plain-language insight	What the reader can do
Recall 0.85, precision 0.89	Of every 100 patients who will be readmitted, we flag 85; 9 in 10 alerts are real.	Plan nurse time for alerts
Predicted probability 0.72	7 out of 10 patients like this one were readmitted.	Prioritise this patient
RMSE $23k	Price estimates are typically within about $23,000.	Set a negotiation range
Top attribution: debt ratio	The main reason for the decision was the debt ratio.	Explain it to the applicant
Cluster 3 (n = 35)	“Careful earners”: 35 high-income, low-spend customers.	Target with a new offer

Rules: use the reader's units (patients, dollars, hours), round sensibly, define any term once, and say what it means for the decision.

Communicating risk and uncertainty

Risk communication presents probabilities and uncertainty in forms non-experts read correctly, such as natural frequencies, absolute numbers and ranges.



15 out of 100 patients like this one

1. **Use counts:** Say “15 out of 100 patients”, not “0.15”.
2. **Give absolute numbers:** “Halves the risk” means “from 2 in 100 to 1 in 100”.
3. **Give a range:** “Between $270k and $335k”, not “$301,482”.

Why: people read counts out of 100 correctly, but often misread decimals and relative changes.

From prediction to action

An actionable insight links a model's prediction to a specific decision, who should take it, and its expected impact.

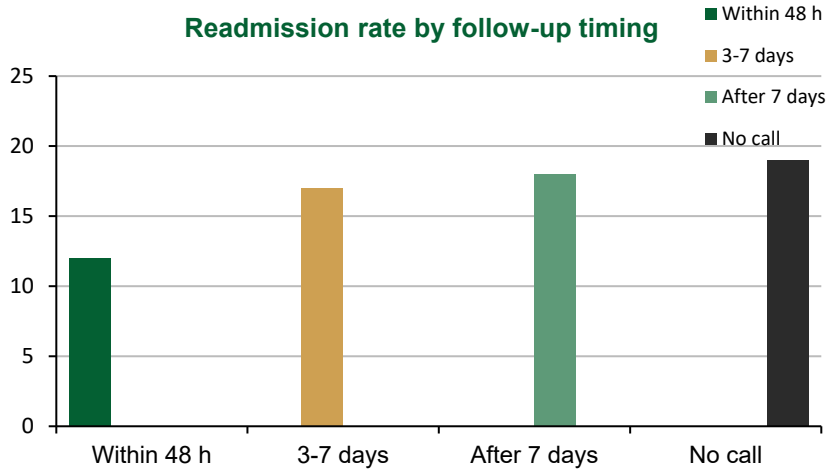
1. **Prediction:** 500 customers flagged as likely to leave.
2. **Insight:** 60% are on month-to-month contracts after a price rise.
3. **Recommendation:** Offer the top 200 a 12-month contract with a discount.
4. **Expected impact:** Costs about $4,000; saves about $8,000 if 1 in 10 stay.

Business numbers	Value
Cost per call and offer	$20
Value of keeping a customer	$400 / year
200 calls cost	$4,000
If 1 in 10 stay (20)	**$8,000 saved**

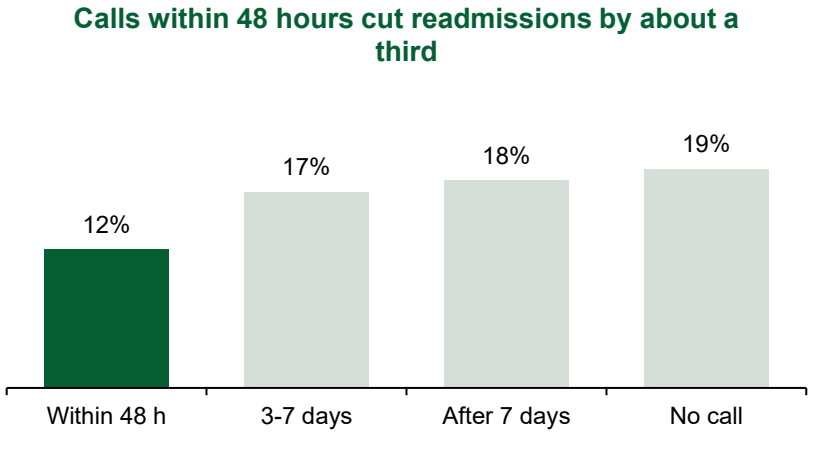
Say it in their currency: dollars, hours or patients, not precision and recall.

Charts that tell the story

An action title states the chart's message in a sentence; decluttering removes everything that does not support that message.



Default: colours, legend, gridlines, descriptive title



Focused: one highlight, direct labels, action title

Rule: every element must earn its place; colour points, it does not decorate.

Ethical visualization

Key question

How do charts mislead, and how do we keep them honest?

What is ethical visualization?

Ethical visualization represents data truthfully, so that viewers reach the conclusion the data actually supports.

1 Proportional

Visual size matches the numbers.

Shown in: Truncated axes; misleading scales

2 Complete

Nothing hidden that changes the conclusion.

Shown in: Selective reporting

3 Contextual

Units, scales and uncertainty shown.

Shown in: Other distortions

4 Accessible

Readable by everyone, including colour-blind viewers.

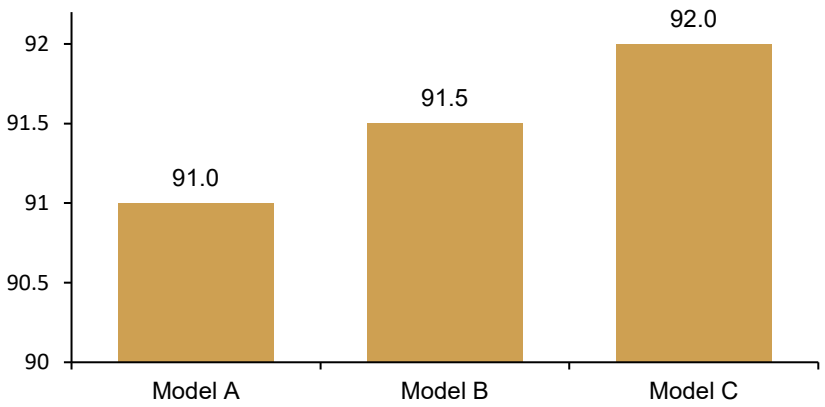
Shown in: Other distortions

Often accidental: software defaults and the wish for a clear message cause most misleading charts; the effect on decisions is the same.

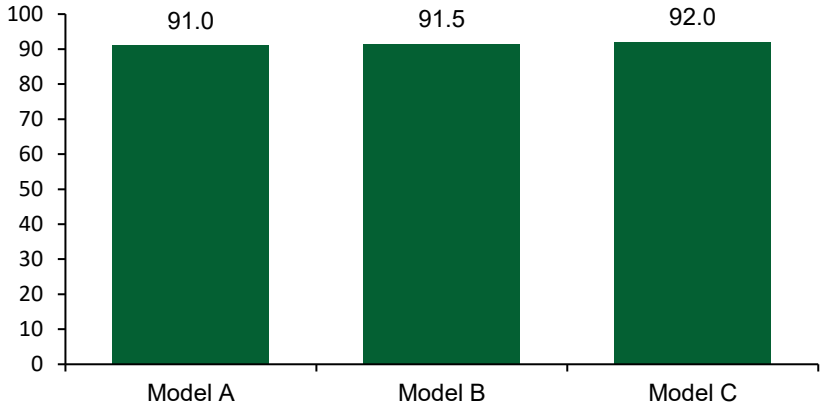
Truncated axes on bar charts

A truncated axis starts above zero. On a bar chart it makes small differences look large, because readers compare bar lengths.

Misleading: axis starts at 90



Honest: axis starts at 0

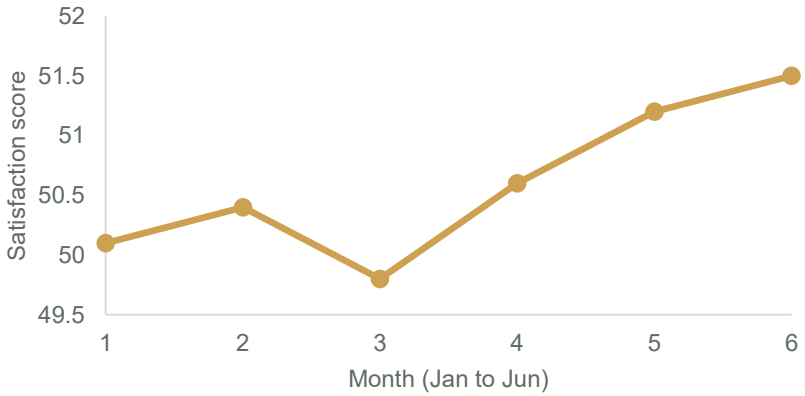


Rule: bar lengths encode values, so bar axes start at zero. For small differences, show the numbers or a labelled dot plot.

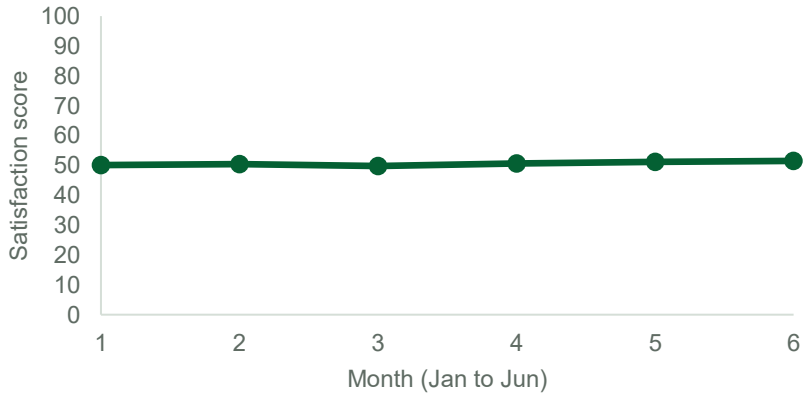
Misleading scales on line charts

A misleading scale is an axis range or type chosen so that a change looks bigger, smaller or steadier than it is.

Zoomed: 49.5 to 52



Full scale: 0 to 100



For line charts: a zoomed axis can be fine if labelled; state the change in words (“up 1.4 points”). Beware dual axes and log scales.

Selective reporting

Selective reporting means showing only the metrics, runs, groups or time periods that support the message, while hiding those that do not.

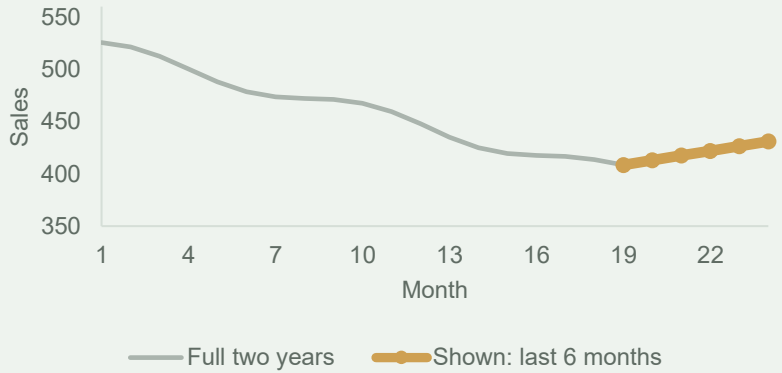
Selective metrics

Headline: “Our fraud model is 99% accurate”

Full picture	Model	Flag nothing
Accuracy	99%	99%
Frauds caught (of 100)	**20**	0

True headline, false impression.

Selective time window



Fix: decide which metrics, runs, groups and time window you will report before looking at the results.

Ethical visualization checklist

An ethical visualization checklist is a short set of questions to answer before a chart is shared.

Check	Question to ask
Scale	Do bars start at zero? Is any zoomed axis clearly labelled?
Completeness	Am I showing all agreed metrics, the baseline, and the full time period?
Uncertainty	Can the reader see how sure we are (ranges, intervals, sample sizes)?
Proportion	Does the visual size of each difference match the numbers?
Accessibility	Readable in greyscale and by colour-blind viewers? Labelled directly?
Message	Does the title state a conclusion the data actually supports?

Fresh-eyes test: ask someone new what the chart says; if it differs from what the data supports, fix it.